Sultan Qaboos University
College of Engineering
Office of Assistant Dean for
Industrial Training and Community Services

Weekly Report
Week #: ~~5~~ 7
Date:

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Tasks Completed (Briefly Describe any tasks that you completed during this week):

We learned electronic components.
We learned how to use multimeter.
We learned the different between sigle phase & three phase

Tasks in Progress (if any):

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Plan for next week (if different from this week):

- Drilling water wells and pumbs.
- Types of submersible and surface pumps.
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Problems/Challenges/Recommendations (if any):

No major problems we phase.

Supervisor Comments (your feedback is very important for our continuing improvement):

No Comments

Student:
I have read and understood the whole content of the Students' Training Manual.

Signature: ....

Supervisor:
I will do my best to guide the student towards achieving all required deliverables.

Signature: ....

Note: A copy of this report should be faxed to 24413416 or scanned and emailed to your SQU supervisor on a weekly basis. Originals should be placed in the Appendix of the **Final Report**.

WEEKLY DEPARTMENT REPORT

Electronics Workshop Operations & Electrical Fundamentals Summary

Department: Electronics & Hardware Engineering

Report Period: Weekly Summary

Prepared By: Electronics Workshop Section Lead

Status: Completed / Reviewed

1. ELECTRONICS WORKSHOP SECTION OVERVIEW

The **Electronics Workshop Section** serves as the core facility for electronic circuit design, testing, prototyping, troubleshooting, and maintenance of modern electrical and electronic equipment. The section is structured to maintain strict safety, quality control, and operational efficiency.

Key Operational Areas

- **Prototyping & Soldering Bench:** Equipped with temperature-controlled soldering irons, hot-air rework stations, and fume extractors.
- **Testing & Measurement Station:** Features oscilloscopes, function generators, regulated DC power supplies, and digital multimeters.
- **Component Inventory & Storage:** Organized storage with anti-static (ESD) protection for sensitive microcontrollers and ICs.

Core Safety & ESD Protocols

- **ESD Protection:** Mandatory use of grounding wrist straps and conductive ESD mats to protect sensitive semiconductors.
- **Electrical Safety:** Isolation transformers and residual current devices (RCDs) installed at main workstations.
- **Ventilation & PPE:** Active exhaust systems during soldering; eye protection mandatory during cutting and grinding.

2. KEY ELECTRONICS COMPONENTS

Electronic components are categorized into **passive** components (which cannot introduce net energy into a circuit) and **active** components (which rely on a source of energy and can control electrical current).

Category	Component	Symbol / Unit	Primary Function & Applications
Passive Components	**Resistor**	R (Ohms, Ω)	Limits electric current flow and divides voltage levels within a circuit.
	Capacitor	C (Farads, F)	Stores electrical energy electrostatically; used for filtering noise, smoothing DC voltage, and signal coupling.
	Inductor	L (Henries, H)	Stores energy in a magnetic field; blocks high-frequency AC signals while allowing DC to pass.
Active Components	**Diode**	D (Anode / Cathode)	Allows electric current to flow in one direction only; used for rectification (converting AC to DC) and circuit protection.
	Transistor (BJT / MOSFET)	Q (Base, Collector, Emitter)	Acts as an electronic switch or signal amplifier; foundational block for microcontrollers and switching circuits.
	Integrated Circuit (IC)	U (Pin configuration)	Miniaturized complex circuits (e.g., Op-Amps, 555 Timers, Microcontrollers) housed on a single semiconductor chip.

3. HOW TO USE A DIGITAL MULTIMETER (DMM)

A Digital Multimeter is an indispensable diagnostic tool used to measure electrical properties such as Voltage (V), Current (I), Resistance (R), and Continuity.

Step-by-Step Operating Procedures

1. Measuring DC/AC Voltage (V):

- Connect the **Black probe** to `COM` and **Red probe** to `V/Ω`.
- Set dial to $V\overline{..}$ (DC Voltage) or $V\sim$ (AC Voltage).
- Connect probes in *parallel* across the component or power source.

2. Measuring Resistance (Ω):

- Ensure power to the circuit is **completely turned off**.
- Set dial to Ω. Place probes across the resistor.

Current Measurement & Continuity

3. Measuring Current (A):

- Move Red probe to `10A` or `mA` port depending on expected load.
- Set dial to $A\overline{..}$ or $A\sim$.
- Break the circuit and place meter in *series* with the load.

4. Continuity Testing ($\ sound\ icon$):

- Select continuity mode. Touch probe tips together (meter should beep).
- Place probes across a trace or wire to check for open/closed loops.

Critical Safety Warning for Current Measurement: Never measure current in parallel with a voltage source. Doing so creates a direct short circuit across the meter's internal low-resistance shunt, blown fuses, or potential damage to the instrument and equipment.

4. SINGLE-PHASE VS. THREE-PHASE POWER SYSTEMS

In electrical power distribution, understanding the distinction between single-phase and three-phase AC systems is essential for proper equipment installation and workshop safety.

Feature / Parameter	Single-Phase System (1-Phase)	Three-Phase System (3-Phase)
Definition	AC power distributed using one active phase wire and one neutral wire.	AC power distributed using three active phase wires (120° apart) and an optional neutral.
Conductors Required	2 Wires (1 Phase + 1 Neutral) (+ Earth)	3 or 4 Wires (3 Phases + 1 Neutral) (+ Earth)
Typical Voltage (Standard)	230V AC (or 120V depending on region)	400V / 415V AC (Line-to-Line)
Waveform & Power Flow	Pulsating power delivery; voltage drops to zero 100/120 times per second.	Continuous and constant power delivery; total sum of voltages never drops to zero.
Efficiency & Capacity	Lower efficiency for heavy loads; requires larger wire gauge for high power.	Higher efficiency; delivers 1.732 times more power than single-phase with less copper.
Common Applications	Residential homes, light commercial loads, small office appliances, basic lab equipment.	Industrial workshops, large electric motors, HVAC compressors, heavy machinery.

Summary Recommendation for Workshop Operations: Standard benchtop tools, soldering irons, and measurement devices operate efficiently on **Single-Phase power**. High-power workshop equipment (e.g., CNC milling machines, air compressors, industrial 3D printers) require a **Three-Phase connection** for balanced loads, lower operating current, and higher operational efficiency.